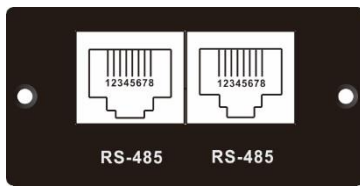


# **BMS Communication Protocol**

## **RS485**

| Version | Details       | Date       | Note |
|---------|---------------|------------|------|
| V01     | First Version | 2021-06-08 |      |
|         |               |            |      |
|         |               |            |      |
|         |               |            |      |

# 1. BMS Pin Definition



|       | Definition |
|-------|------------|
| PIN 5 | RS485B     |
| PIN 6 | RS485A     |

## 2. Communication parameter configuration

| Baud rate | Start bit | Data bit | Parity bit | Stop bit |
|-----------|-----------|----------|------------|----------|
| 9600      | 1         | 8        | N          | 1        |

## 3. Communication frame format

### 3.1 Version information inquiry

| Query                           | Hex | Response        | Hex | Data Note            | Unit |
|---------------------------------|-----|-----------------|-----|----------------------|------|
| Slave ID                        | 01  | Slave ID        | 01  |                      |      |
| Command type                    | 03  | Command type    | 03  |                      |      |
| Start Address of data high byte | F8  | Data Length * 2 | 02  |                      |      |
| Start Address of data low byte  | 0B  | Data1 high byte | 00  | BMS firmware version |      |
| Data Length high byte           | 00  | Data1 low byte  | 15  |                      |      |
| Data Length low byte            | 01  | CRC low byte    | 79  |                      |      |
| CRC low byte                    | C4  | CRC high byte   | 8B  |                      |      |
| CRC high byte                   | A8  |                 |     |                      |      |

### 3.2 Cell voltage and temperature information inquiry

| Query                           | Hex | Response        | Hex | Data Note     | Unit |
|---------------------------------|-----|-----------------|-----|---------------|------|
| Slave ID                        | 01  | Slave ID        | 01  |               |      |
| Command type                    | 03  | Command type    | 03  |               |      |
| Start Address of data high byte | 13  | Data Length * 2 | 30  |               |      |
| Start Address of data low byte  | 2A  | Data1 high byte | 0C  | Cell1 voltage | 1mV  |
| Data Length high byte           | 00  | Data1 low byte  | 80  |               |      |
| Data Length low byte            | 18  | Data2 high byte | 0C  | Cell2 voltage | 1mV  |
| CRC low byte                    | 60  | Data2 low byte  | 80  |               |      |
| CRC high byte                   | 8C  | Data3 high byte | 0C  | Cell3 voltage | 1mV  |
|                                 |     | Data3 low byte  | 80  |               |      |
|                                 |     | Data4 high byte | 0C  | Cell4 voltage | 1mV  |

|  |  |                  |    |                      |     |
|--|--|------------------|----|----------------------|-----|
|  |  | Data4 low byte   | 80 |                      |     |
|  |  | Data5 high byte  | 0C | Cell5 voltage        | 1mV |
|  |  | Data5 low byte   | 80 |                      |     |
|  |  | Data6 high byte  | 0C | Cell6 voltage        | 1mV |
|  |  | Data6 low byte   | 80 |                      |     |
|  |  | Data7 high byte  | 0C | Cell7 voltage        | 1mV |
|  |  | Data7 low byte   | 80 |                      |     |
|  |  | Data8 high byte  | 0C | Cell8 voltage        | 1mV |
|  |  | Data8 low byte   | 80 |                      |     |
|  |  | Data9 high byte  | 0C | Cell9 voltage        | 1mV |
|  |  | Data9 low byte   | 80 |                      |     |
|  |  | Data10 high byte | 0C | Cell10 voltage       | 1mV |
|  |  | Data10 low byte  | 80 |                      |     |
|  |  | Data11 high byte | 0C | Cell11 voltage       | 1mV |
|  |  | Data11 low byte  | 80 |                      |     |
|  |  | Data12 high byte | 0C | Cell12 voltage       | 1mV |
|  |  | Data12 low byte  | 80 |                      |     |
|  |  | Data13 high byte | 0C | Cell13 voltage       | 1mV |
|  |  | Data13 low byte  | 80 |                      |     |
|  |  | Data14 high byte | 0C | Cell14 voltage       | 1mV |
|  |  | Data14 low byte  | 80 |                      |     |
|  |  | Data15 high byte | 0C | Cell15 voltage       | 1mV |
|  |  | Data15 low byte  | 80 |                      |     |
|  |  | Data16 high byte | 0C | Cell16 voltage       | 1mV |
|  |  | Data16 low byte  | 80 |                      |     |
|  |  | Data17 high byte | 00 | Temperature Sensor 1 | 1°C |
|  |  | Data17 low byte  | 19 |                      |     |
|  |  | Data18 high byte | 00 | Temperature Sensor 2 | 1°C |
|  |  | Data18 low byte  | 19 |                      |     |
|  |  | Data19 high byte | 00 | Temperature Sensor 3 | 1°C |
|  |  | Data19 low byte  | 19 |                      |     |
|  |  | Data20 high byte | 00 | Temperature Sensor 4 | 1°C |
|  |  | Data20 low byte  | 19 |                      |     |
|  |  | Data21 high byte | 00 | Temperature Sensor 5 | 1°C |
|  |  | Data21 low byte  | 19 |                      |     |
|  |  | Data22 high byte | 00 | Temperature Sensor 6 | 1°C |
|  |  | Data22 low byte  | 19 |                      |     |
|  |  | Data23 high byte | 00 | Temperature Sensor 7 | 1°C |
|  |  | Data23 low byte  | 19 |                      |     |
|  |  | Data24 high byte | 00 | Temperature Sensor 8 | 1°C |
|  |  | Data24 low byte  | 19 |                      |     |
|  |  | CRC low byte     | 03 |                      |     |
|  |  | CRC high byte    | C7 |                      |     |

### 3.3 Charger and discharge information inquiry

| Query                           | Hex | Response        | Hex | Data Note               | Unit  |
|---------------------------------|-----|-----------------|-----|-------------------------|-------|
| Slave ID                        | 01  | Slave ID        | 01  |                         |       |
| Command type                    | 03  | Command type    | 03  |                         |       |
| Start Address of data high byte | 13  | Data Length * 2 | 08  |                         |       |
| Start Address of data low byte  | 1C  | Data1 high byte | 16  | Charge voltage limit    | 0.01V |
| Data Length high byte           | 00  | Data1 low byte  | 80  |                         |       |
| Data Length low byte            | 04  | Data2 high byte | 12  | Discharge voltage limit | 0.01V |
| CRC low byte                    | 81  | Data2 low byte  | C0  |                         |       |
| CRC high byte                   | 4B  | Data3 high byte | 02  | Charge current limit    | 0.1A  |
|                                 |     | Data3 low byte  | 58  |                         |       |
|                                 |     | Data4 high byte | 02  | Discharge current limit | 0.1A  |
|                                 |     | Data4 low byte  | 58  |                         |       |
|                                 |     | CRC low byte    | 16  |                         |       |
|                                 |     | CRC high byte   | AB  |                         |       |

### 3.4 Battery information inquiry

| Query                           | Hex | Response         | Hex | Note            | Unit  |
|---------------------------------|-----|------------------|-----|-----------------|-------|
| Slave ID                        | 01  | Slave ID         | 01  |                 |       |
| Command type                    | 03  | Command type     | 03  |                 |       |
| Start Address of data high byte | 13  | Data Length * 2  | 14  |                 |       |
| Start Address of data low byte  | 02  | Data1 high byte  | 00  | Battery Status  |       |
| Data Length high byte           | 00  | Data1 low byte   | 05  |                 |       |
| Data Length low byte            | 0A  | Data2 high byte  | 00  | /               |       |
| CRC low byte                    | 60  | Data2 low byte   | 00  |                 |       |
| CRC high byte                   | 89  | Data3 high byte  | 00  | Fault Status    |       |
|                                 |     | Data3 low byte   | 00  |                 |       |
|                                 |     | Data4 high byte  | 00  | /               |       |
|                                 |     | Data4 low byte   | 00  |                 |       |
|                                 |     | Data5 high byte  | 14  | Battery Voltage | 0.01V |
|                                 |     | Data5 low byte   | 00  |                 |       |
|                                 |     | Data6 high byte  | 00  | Battery Current | 0.1A  |
|                                 |     | Data6 low byte   | 00  |                 |       |
|                                 |     | Data7 high byte  | 00  | /               |       |
|                                 |     | Data7 low byte   | 00  |                 |       |
|                                 |     | Data8 high byte  | 00  | /               |       |
|                                 |     | Data8 low byte   | 00  |                 |       |
|                                 |     | Data9 high byte  | 00  | BMS Temperature | 1℃    |
|                                 |     | Data9 low byte   | 19  |                 |       |
|                                 |     | Data10 high byte | 00  | SOC             | %     |
|                                 |     | Data10 low byte  | 50  |                 |       |
|                                 |     | CRC low byte     | 8E  |                 |       |
|                                 |     | CRC high byte    | 60  |                 |       |

\* The CRC check range is all of the bytes before the CRC field.

\* Battery Current

Charging current is negative and discharging current is positive.

\* Battery Status

| Bit  | Content            | Note                                |
|------|--------------------|-------------------------------------|
| 0    | Charge enable      | 1: yes<br>0: request stop charge    |
| 1    | Charge immediately | 1: request<br>0: no request         |
| 2    | Discharge enable   | 1: yes<br>0: request stop discharge |
| 3-15 | Reserved           |                                     |

\* Fault Status

| Bit   | Content                       | Note                     |
|-------|-------------------------------|--------------------------|
| 0     | Reserved                      |                          |
| 1     | Reserved                      |                          |
| 2     | Cell voltage high status      | 0: normal<br>1: abnormal |
| 3     | Cell voltage low status       | 0: normal<br>1: abnormal |
| 4     | Charge current high status    | 0: normal<br>1: abnormal |
| 5     | Discharge current high status | 0: normal<br>1: abnormal |
| 6     | BMS Temperature high status   | 0: normal<br>1: abnormal |
| 7     | Reserved                      | 0: normal<br>1: abnormal |
| 8     | Cell Temperature high status  | 0: normal<br>1: abnormal |
| 9     | Cell Temperature low status   | 0: normal<br>1: abnormal |
| 10-15 | Reserved                      |                          |